

SESSION WILL BE DIVIDED INTO

1. What is meant by M Functions / M Language?
2. Why do we need it?
3. Syntax of M Language
4. Benefits of Knowing M Functions
5. Real Life Saviours

SECTION 1

WHAT IS M Language ?

- M language or M code is nothing but the formula language that **drives the data handling and transformations in Power BI.**
- The M in M language stands for Mashup.
- M language is used in the Power Query Editor.

SECTION 1

WHAT IS M Language ?

- Total Number of Functions in M language?

The Power Query M function reference includes articles for each of the **over 700** functions. These reference articles are auto-generated from in-product help. 20 Sept 2022



Microsoft Learn

<https://learn.microsoft.com> › Learn › Power Query M ›

Power Query M function reference - Microsoft Learn

 About featured snippets •  Feedback

- Microsoft PDF Link: [PDF Link](#)

SECTION 1

WHAT IS M Language ?

- BEST PART ABOUT M CODE.
 - Majority of the work is done by Power BI in Power Query Editor.
 - No need to remember every formula.
 - UI handles the majority of the work.

GOAL OF THE SESSION

TO UNDERSTAND THE MEANING OF M CODE,
RATHER THAN JUST COPYING AND PASTING IT.
GIVE YOU MY RESOURCES FROM EXPERIENCE.

SECTION 2

WHY DO WE NEED IT?

- Working of the Power Query Editor.
- What is happening in the background

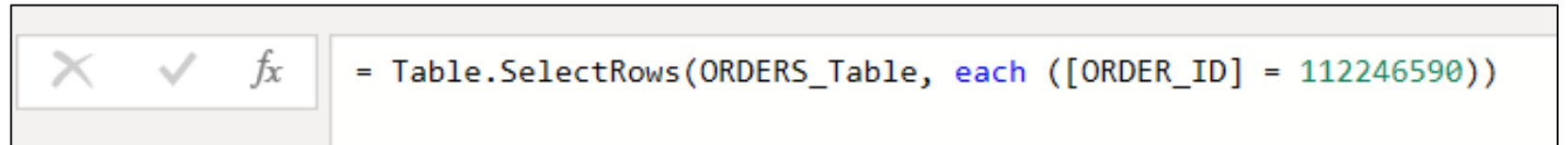
The screenshot displays the Power Query Editor interface. At the top, the formula bar shows the query definition: `= Table.Sort(ORDERS_Table,{{"DATE_", Order.Ascending}})`. Below this, a table of data is visible with columns: **DATE**, **CITY_NAME**, **ORDER_ID**, **CART_ID**, **DIM_CUSTOMER_KEY**, **PROCURED_QUANTITY**, and **UNIT_SELLIN**. A 'Sort Ascending' dialog box is open over the table, showing options to 'Clear Sort', 'Clear Filter', 'Remove Empty', and 'Date Filters'. The 'Date Filters' section includes a search bar and a list of filters: '(Select All)' and '01-04-2022'. The 'APPLIED STEPS' pane on the right shows the sequence of steps: 'Source', 'Navigation', and 'Sorted Rows'. The 'Sorted Rows' step is highlighted with a blue box. The bottom status bar indicates '10 COLUMNS, 999+ ROWS' and 'Column profiling based on top 1000 rows'.

DATE	CITY_NAME	ORDER_ID	CART_ID	DIM_CUSTOMER_KEY	PROCURED_QUANTITY	UNIT_SELLIN
01-04-2022	Mumbai	112247182	173273662	16541671	1	
01-04-2022	Bengaluru	112246976	173273597	18259433	1	
01-04-2022	HR-NCR	112247149	173274225	479129	1	
01-04-2022	Delhi	112247317	173274278	13497358	1	
01-04-2022	Delhi	112247233	173269263	21466	1	
01-04-2022	HR-NCR	112247321	173273059	4480518	1	
01-04-2022	Delhi	112247317	173274278	13497358	1	
01-04-2022	Bengaluru	112247425	173273350	17213227	1	
01-04-2022	Bengaluru	112247328	173269880	18049405	2	
01-04-2022	HR-NCR	112247431	173274695	1778240	1	
01-04-2022	HR-NCR	112247464	173273818	13982586	1	
01-04-2022	Delhi	112247443	169126739	7258345	1	
01-04-2022	Bengaluru	112247487	173275066	2812501	2	
01-04-2022	Delhi	112247483	173274915	5792943	1	
01-04-2022	Bengaluru	112247487	173275066	2812501	1	
01-04-2022	Delhi	112247513	172624184	18062502	2	
01-04-2022	Delhi	112247522	173275012	17550576	1	
01-04-2022	HR-NCR	112247528	173117975	2394931	1	
01-04-2022	HR-NCR	112247536	173274618	5507569	1	
01-04-2022	Delhi	112247584	172746884	2046368	1	
01-04-2022	Delhi	112247638	173274990	16407400	1	
01-04-2022	HR-NCR	112247693	173273354	18275433	1	
01-04-2022	Delhi	112247673	151098842	7547491	1	
01-04-2022	Delhi	112247712	173275226	2973152	1	
01-04-2022	HR-NCR	112247731	173271251	2134324	1	
01-04-2022	Bengaluru	112247750	173274184	16712007	2	
01-04-2022	Delhi	112247744	173275338	6981521	1	

SECTION 3

M Language Syntax

There is a defined syntax that we need to stick to for using M language.

A screenshot of a software interface, likely Power BI, showing a formula bar. The bar has a light gray background and a thin border. On the left side of the bar, there are three small icons: a red 'X' for cancel, a green checkmark for confirm, and a blue 'fx' for functions. To the right of these icons, the formula is entered: `= Table.SelectRows(ORDERS_Table, each ([ORDER_ID] = 112246590))`. The text is in a monospaced font, with some words like 'each' and the number '112246590' highlighted in blue and green respectively.

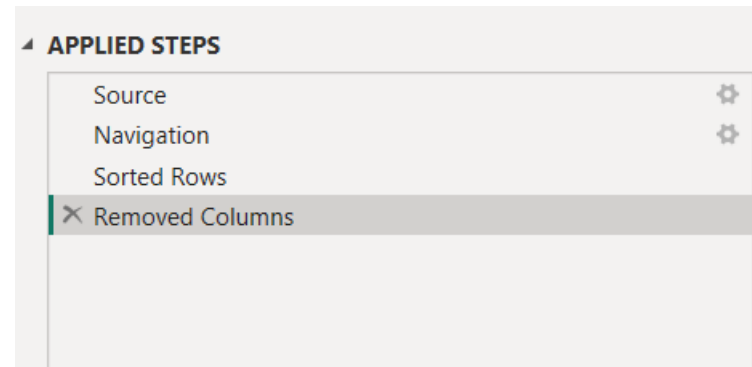
```
= Table.SelectRows(ORDERS_Table, each ([ORDER_ID] = 112246590))
```

1. Table.SelectRows – Function Name
2. ORDERS_Table – Reference to previous step.
This argument allows Power BI to understand the previous applied step.
3. Each ([ORDER_ID] = 112246590) – Argument of the function.

SECTION 4

Benefits of knowing M-Code.

- Editing of the code.

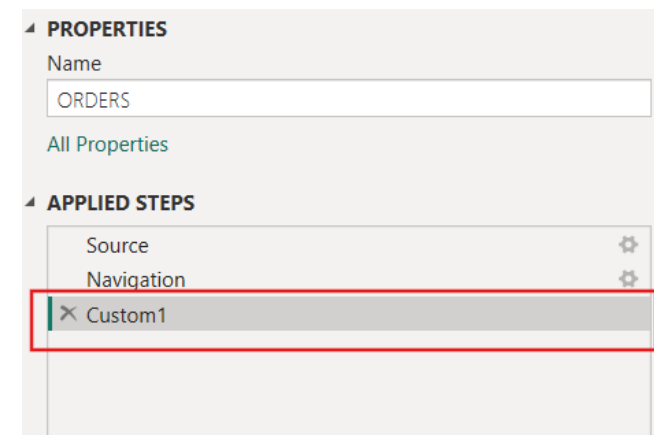
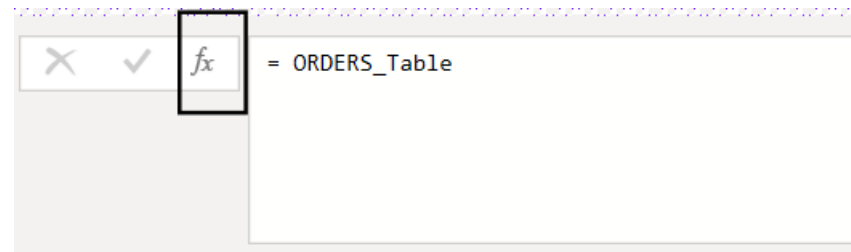


```
= Table.RemoveColumns("#Sorted Rows",{ "CITY_NAME", "ORDER_ID", "CART_ID", "DIM_CUSTOMER_KEY", "PROCURED_QUANTITY",  
"UNIT_SELLING_PRICE", "TOTAL_DISCOUNT_AMOUNT", "PRODUCT_ID", "TOTAL_WEIGHTED_LANDING_PRICE"})
```


SECTION 4

Benefits of knowing M-Code.

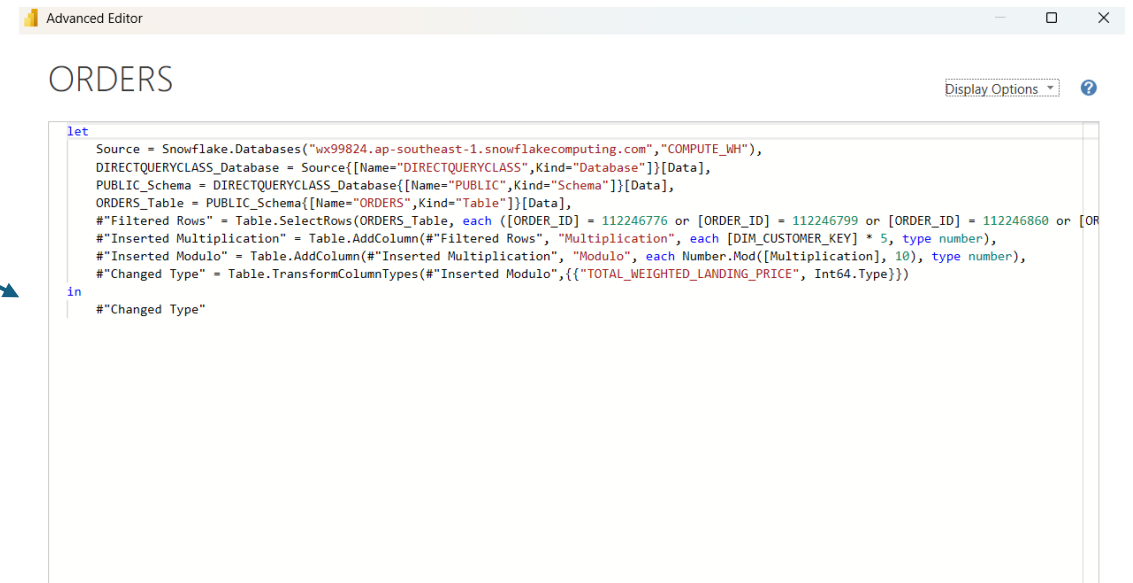
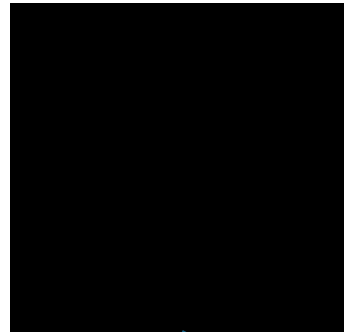
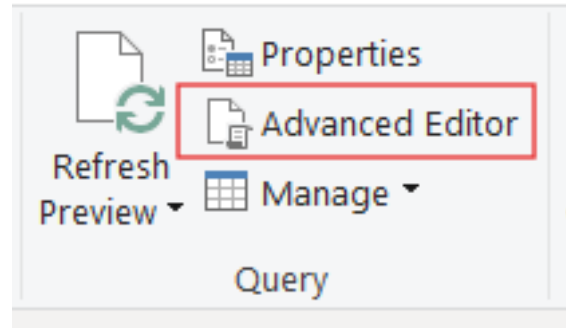
- Making new codes.



SECTION 4

Benefits of knowing M-Code.

- Advance Editor Understanding.

A screenshot of the 'Advanced Editor' window in Snowflake. The window title is 'Advanced Editor'. The main content area is titled 'ORDERS'. Below the title, there is a 'Display Options' dropdown and a help icon. The SQL query is displayed in a monospace font with syntax highlighting. The query starts with 'let' and defines several variables: 'Source', 'DIRECTQUERYCLASS_Database', 'PUBLIC_Schema', and 'ORDERS_Table'. It then uses 'Table.SelectRows' to filter data based on 'ORDER_ID' and 'Table.AddColumn' to add calculated columns for 'Filtered Rows', 'Inserted Multiplication', 'Inserted Modulo', and 'Changed Type'. The query ends with 'in' and another 'Changed Type' statement.

```
let
Source = Snowflake.Databases("wx99824.ap-southeast-1.snowflakecomputing.com", "COMPUTE_WH"),
DIRECTQUERYCLASS_Database = Source[{Name="DIRECTQUERYCLASS", Kind="Database"}][Data],
PUBLIC_Schema = DIRECTQUERYCLASS_Database[{Name="PUBLIC", Kind="Schema"}][Data],
ORDERS_Table = PUBLIC_Schema[{Name="ORDERS", Kind="Table"}][Data],
#"Filtered Rows" = Table.SelectRows(ORDERS_Table, each ([ORDER_ID] = 112246776 or [ORDER_ID] = 112246799 or [ORDER_ID] = 112246860 or [OR
#"Inserted Multiplication" = Table.AddColumn(#"Filtered Rows", "Multiplication", each [DIM_CUSTOMER_KEY] * 5, type number),
#"Inserted Modulo" = Table.AddColumn(#"Inserted Multiplication", "Modulo", each Number.Mod([Multiplication], 10), type number),
#"Changed Type" = Table.TransformColumnTypes(#"Inserted Modulo",{{"TOTAL_WEIGHTED_LANDING_PRICE", Int64.Type}})
in
#"Changed Type"
```

SECTION 4

Benefits of knowing M-Code.

- Advance Editor Understanding.

Let and in keyword – COMPULSORY TO USE

let Keyword:

The let keyword is used to declare variables within an M language script. Variables declared using let are scoped to the expression following the in keyword.

You can declare multiple variables within a single let statement, separated by commas.

in Keyword:

The in keyword is used to specify the expression in which the declared variables will be used.